Library-Style Product Catalogue Audit (Case Study 49)

Problem Statement

Store product ID, price, and stock in tuples. Calculate stock value and identify products with low stock.

Objectives

Practise tuples, tuple unpacking, loops, and conditions.

Assumptions

A product tuple is (product_id, price, quantity). Low stock means fewer than five items.

Algorithm

1. Store product records as tuples. 2. Unpack each tuple. 3. Calculate price multiplied by quantity. 4. Print each product's
stock value and low-stock warning.

Flowchart (Text Form)

START → Create tuple records → Unpack one record → Calculate stock value → Check stock < 5 → Print report →
STOP.

Pseudocode

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```

Python Source Code

```
products = [
    ("P101", 50, 10),
    ("P102", 120, 3),
    ("P103", 75, 7)
]
total_value = 0
for product_id, price, quantity in products:
    value = price * quantity
    total_value += value
    print(product_id, "stock value: Rs.", value)
    if quantity &lt; 5:
        print("Low stock:", product_id)
print("Total inventory value: Rs.", total_value)
```

Test Case and Expected Output

Product P102 has price 120 and quantity 3
Expected: P102 stock value Rs. 360; Low stock: P102; Total inventory value
Rs. 1535

Conclusion

The program uses tuples and unpacking to audit stock quantities and calculate inventory value.